

Malcolm Pereira

Data Engineer - Data Scientist - Web Dev

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Profile

Multi-disciplined Data Engineer with experience in banking digital transformation, specialising in **data pipelines, automation, analytics, and machine learning applications**. Skilled in **Python, SQL, web scraping, and data visualisation**, with experience delivering internal tools and automation workflows for business teams. Currently developing a web store with **PERN** stack, and undergoing Graduate Training for **AWS re/Start** programme. Seeking opportunities in **Data Engineering, Data Science, Web Dev or AI/ML roles**.

Public Portfolio Projects

- Malaysian Banking YouTube Analysis** 
- Built a complete end-to-end dashboard combining **web scraping, database management, sentiment analysis, LLM API to create a Text-to-SQL AI Query Assistant App** for analysing banking-related YouTube content.
- PDF Table Extraction Engine** 
- Developed an **OCR-based pipeline using OpenCV and Tesseract** to convert scanned(image-based) PDFs into structured tabular data.
- Fantasyland Probability Simulator** 
- Used **Monte Carlo simulation techniques** to analyse probabilities in Open Face Chinese Poker.
- Full-Stack PERN ToDo App** 
- Built a fully functional ToDo App as a Proof of Concept (PoC)with **PERN** stack (Postgresql, Express, Node.js and React) stack.
 - Integrated **Tailwind**, vanilla and **modular CSS**.
 - Custom **backend logic** to track user interactions in **audit tables**

Work Projects & BAU

- Data Engineering**
- Developed Big Data jobs using Talend Data Studio using components and connectors.
- Legacy Data Recovery (RCRM)**
- Converted legacy **.xls archives into structured CSV datasets**, improving long-term data accessibility and standardising archives.
- Government Tender Data Scraper**
- Developed a **web scraping pipeline** to collect Malaysian government tender data for the Business Banking team.
 - Automated reporting workflow using **Microsoft PowerAutomate** for distribution of insights.
- RPA Automation Improvements**
- Supported the RPA team by implementing **bug fixes and improvements in UiPath automation workflows**.
- ER Diagram Visualisation Tool**
- Built an interactive **ER diagram dashboard** using Python, Plotly Dash, and NetworkX to visualise database relationships for internal data engineering workflows.
- Gantt Chart Task Monitor PMO**
- Developed a data pipeline converting **email notifications into Gantt chart visualisations** for *monitoring progress of UAT and NF-UAT activities*. 

Education & Certificates

- Certified Artificial Intelligence Engineer (CAIE)**
SG Academy 
Jul 2026 – Dec 2026
USAII — Sponsored Programme
- Bachelor of Mechanical Engineering (BEng), 2:1 Honours**
Queen Mary University of London
Sep 2017 – May 2021
Treasurer – QMUL Chess Society
- Managed society finances and membership growth from **40 to 52 members** through events and marketing.

Professional Experience

- Digital Executive / Data Engineer**
AmBank Group
Apr 2023 – Present | Klang Valley, Malaysia
- Rotated across **Digitalisation (RPA), Group Data Science & Analytics (GDSA), and Retail Recovery Management (RCRM)** teams.
 - Built internal automation tools to streamline operational workflows and data reporting.
 - Analysed customer repayment behaviour to improve **SMS communication strategies** within recovery operations.
 - Monitored and maintained production data ingestion pipelines into the organisation's **data lake**, resolving pipeline issues within daily SLA.
- Coding Instructor (Part-Time, Remote)**
SG Academy
Dec 2024 – May 2025
- Taught beginner-level Python programming and computational thinking to multiple student cohorts.
 - Delivered lessons using **interactive Jupyter notebooks and dynamic lecture slides**.
 - Developed reusable teaching materials to standardise curriculum delivery.
 - Automated environment creation setup for students in **Powershell and Bash**, facilitating classroom delivery.
 - Maintained a **3.5/5 student satisfaction rating**.

Skills

- Programming** ● ● ● ● ● ●
- Python** (advanced), SQL, Bash scripting, **openpyxl**
 - Strong experience with **modular design**, OOP, PEP-8 standards, and **virtual environments**
- Data Science & Analytics** ● ● ● ● ● ●
- pandas, numpy, matplotlib, seaborn, plotly-dash**
 - Exploratory Data Analysis (**EDA**), data cleaning, feature engineering
 - Monte Carlo simulations and statistical modelling
 - Visualisations with PowerBI, **DQ rule** compliance
- Databases** ● ● ● ● ● ●
- MySQL, PostgreSQL, SQLite
 - Database schema design, ER diagrams, **normalisation**, NoSQL with MongoDB
 - Graph databases** (Neo4j, Cypher, networkx)
- Machine Learning & AI** ● ● ● ● ● ●
- scikit-learn (**Logistic Regression**, KNN, Decision Trees, SVM)
 - ML pipelines, model evaluation, hyperparameter tuning, AI Theory
- Automation & Web Scraping** ● ● ● ● ● ●
- Selenium, BeautifulSoup, **crontab**
 - Automation workflows** for data collection and reporting.
- Unstructured Data Processing** ● ● ● ● ● ●
- NLP** and sentiment analysis
 - OCR** pipelines for scanned document **table extraction**
- Web Dev & DevOps** ● ● ● ● ● ●
- Solo & Collaborative Git**, GitHub, Jupyter, VS Code
 - Full Stack **PERN**
 - Linux command line, awk, grep, **regex**